

Heat & Thermodynamics

Postulates of the Kinetic Theory of Gases:

- The molecules in a gas are small and very far apart. Most of the volume which a gas occupies is empty space.
- Gas molecules are in constant random motion. Just as many molecules are moving in one direction as in any other.
- Molecules can collide with each other and with the walls of the container. Collisions with the walls account for the pressure of the gas.
- When collisions occur, the molecules lose no kinetic energy; that is, the collisions are said to be **perfectly elastic**.
- The molecules exert no attractive or repulsive forces on one another except during the process of collision. Between collisions, they move in straight lines.

Mean free path: We know from the kinetic theory of gases that molecules of gas contained in a vessel move randomly in all directions. They collide with each other and also with the walls of the vessel. Between two successive collisions the molecules are travel in straight line. The distance travelled between two collisions is called Free Path. The average distance travelled by a molecule between successive collisions is called *Mean Free Path*. It is denoted by λ . Mathematically-

$$\lambda = \frac{\text{Total distance travelled (S)}}{\text{No of collisions (N)}}$$

Derivation: Clausius derived the relation of mean free path in terms of number of molecules in gas and diameter of a molecule. In deriving this relation he considered that only one molecule is in motion and other molecules in the vessel are in rest.

Let the number of molecules per unit volume of a gas be n and the diameter of each of them is a . Furthermore, let a molecule move with velocity v . Let us draw a circle taking ' a ' as the radius. Let us consider a cylinder of length v having this radius, a .

Then the volume of the cylinder = $\pi a^2 v$

Consider that the said molecule will collide in one second with all the molecules whose centres are contained within the volume of the cylinder.

So, the total number of molecules contained within the cylinder, $N = \pi a^2 v n$

$$\begin{aligned} \text{Hence average distance between two collisions} &= \frac{\text{Total distance travelled (S)}}{\text{No of collisions (N)}} \\ &= \frac{v}{\pi a^2 v n} \end{aligned}$$

$$\therefore \text{Mean Free Path, } \lambda = \frac{1}{\pi a^2 n} \quad \text{----- (1)}$$

If the mass of the each molecule is m , then,

$mn = \text{mass of the gas molecules per unit volume} = \text{density of gas}, \rho$

$$\text{Now, } \lambda = \frac{m}{\pi a^2 mn}$$

$$\lambda = \frac{m}{\pi a^2 \rho}$$

Scientist Clausius, in deriving the above relation, considered that only one molecule is in motion and all other molecules are at rest. But actually all molecules are in random motion. Later on, Maxwell, using his distribution law of velocity, derived the following relation for the mean free path,

$$\lambda = \frac{1}{\sqrt{2} \pi a^2 n}, \quad \text{----- (3)}$$

Generally, this equation is used in calculating mean free path.

Problem-1: The diameter of the molecule of a gas is $3 \times 10^{-10} \text{ m}$ and number of molecules per cm^3 is 6×10^{20} . Calculate the mean free path of the molecules.

Problem-2: The mean free path of the molecules of a gas is $6 \times 10^{-8} \text{ m}$ and the diameter of a molecule is $2.5 \times 10^{-10} \text{ m}$. Find the number of molecule per m^3 .

Equipartition of energy: In physics, a *degree of freedom* is an independent physical parameter in the formal description of the state of a physical system. The set of all dimensions of a system is known as a phase space, and degrees of freedom are sometimes referred to as its dimensions. For a rotational body, the degree of freedom is *five* and for linear motional body, the degree of freedom is *Three*.

The theorem of equipartition of energy states that molecules in thermal equilibrium have the same average energy associated with each independent degree of freedom of their motion and that the energy is $\frac{1}{2} kT$ (Per molecule) or $\frac{1}{2} RT$ (Per mole).

The total thermal energy of system with n degree of freedom $= \frac{n}{2} kT$ or $\frac{n}{2} RT$

(Where, $k = \text{Boltzmann's Constant}$ and $R = \text{Gas Constant}$)

Brownian motion: Brownian motion or pedesis is the random motion of particles suspended in a fluid (a liquid or a gas) resulting from their collision with the quick atoms or molecules in the gas or liquid.

If a number of particles subject to Brownian motion are present in a given medium and there is no preferred direction for the random oscillations, then over a period of time the particles will tend to be spread evenly throughout the medium.